MIL-T-48312(PA)
30 September 1974
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

TEST SET, ATOMIC WEAPON: T-4133 AND T-4162

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test sets entitled Test Set, Atomic Weapon: T-4133 and T-4162 which are used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For.

DRAWINGS

ORDNANCE CORPS

8808362	- Container, Packing, Ammunition, for Domestic (CONUS) Shipments of Special Weapons Items, Packing and Marking.
8813533	- Schematic diagram, Control Sub-Assembly Prefire T-4133.
8821412	- Test Set, Atomic Weapon, T-4133.
8849096	- Test Set, Atomic Weapon, Training, T-4162.
E1-8842555	- Equipment List.

FSC 1190

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test set shall comply with all requirements specified on the drawings (dwgs) 8821412 and 8849096, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Starting torque. The starting torque shall not exceed 25 inch pounds and the average of 5 trials shall not exceed 20 inch pounds when tested in accordance with 4.4.1.

3.6 Insulation resistance. The insulation resistance between pins specified in Table I at temperatures of minus (-) 65 degrees Fahrenheit (°F) plus or minus (±) 5°F and 80 ± 20°F shall be not less than 20 megohms and at 165 ± 5°F shall be not less than 1 megohm when tested at 500 ± 50 volts direct current (vdc). Testing shall be as specified in 4.4.2.

Table I

From Pin	To Pin
P1-8	P1-15, 14, 13, 12, 11, 9, case

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3.7 Continuity. The resistance between the pins specified in Table II shall not exceed 0.070 ohms when tested as specified in 4.4.3.

Table II

From Pin	To Pin
P1-1	P1-2
P1-3	P1-4, P1-6, P1-5
P1-7	P1-8

3.8 Performance. The test set shall provide for detection of open circuits, grounded circuits and a lamp test when tested as specified in 4.4.4. During this test the indicator lamps shall operate as indicated in Table III.

Table III

Line	Position	Gen Rotation	Jumpers From	Jumpers To	Lamps Red	Lamps Green
1	5	ccw*	--	--	ON	OFF
2	6	ccw	P1-3	P1-8	OFF	OFF
3	7	ccw	P1-3	P1-6	ON	ON
4	8	ccw	P1-6	P1-3	OFF	OFF
5	9	cw**	--	--	ON	OFF
6	10	cw	P1-3	P1-8	ON	OFF
7	11	cw	P1-6	P1-3	ON	OFF
8	12	cw	P1-3	P1-1,P1-6,P1-7	ON	OFF
9	13	cw	P1-3	P1-1, P1-8	ON	OFF
10	14	cw	P1-3	P1-1	OFF	ON
11	15	cw	P1-4	P1-2	ON	OFF
12	16	cw	P1-4	P1-2	ON	OFF

*Counterclockwise
**Clockwise

3.9 Diode current leakage. The leakage current of diode CR1, checked at 350 ± 25 vdc between the pins and at polarities listed in Table IV shall not exceed 20 microamperes maximum (max) when tested in accordance with 4.4.5.

Table IV

(For circuit reference see applicable dwg)

Plus (Pin)	Minus (Pin)	Diode Tested
P1-7	P1-1 Leakage	CR1

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3.10 Life. The test set shall have a minimum life span of 600 cycles and shall be capable of meeting the requirements of 3.5 through 3.8 after completion of testing. The total number of cycles shall be distributed over the period of environmental testing as indicated in Table VI. The test shall be conducted as specified in 4.4.6.

3.11 Environmental testing. The test set shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The test set shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.7. Tests shall be conducted in the order specified.

3.11.1 Transportation vibration. The test set shall be vibrated at the frequencies, gravitational units (g) accelerations, temperatures, and for the time specified in Table V, in each of three mutually perpendicular axes. The assembly shall show no evidence of damage such as cracks, chips, separations, or loose parts, and shall comply with the requirements of 3.5 through 3.8 after being tested as specified in 4.4.7.1.

Table V

Frequency Range Hertz (Hz)	Double Displacement or Peak g Level Tolerance + 5 Percent	Cycle Time To Cover Frequency Range (minutes)	Test Time in Minutes Minimum Time Per Axis 80 + 20°F	160 + 5°F	-65 + 5°F
10-20-10	2 g	3 ± 0.5	15	10	10
20-75-20	0.036 inch	3 ± 0.5	15	10	10
75-100-75	5 g	3 ± 0.5	15	10	10

3.11.2 Acceleration. The test set shall withstand acceleration of 40 ± 5 g's for 5 seconds in the direction of forward motion, (see Figure 1) and of 18 ± 1 g for 15 seconds perpendicular to the direction of forward motion. The assembly shall remain mechanically intact during and after testing, but it is not required to function after this test. The test shall be conducted as specified in 4.4.7.2.

3.11.3 Transportation shock. The test set shall be capable of withstanding shocks of 75 ± 8 g's peak amplitude with a half amplitude duration of 9 ± 3 milliseconds in each of three mutually perpendicular axes. The assembly shall comply with the requirements of 3.5 through 3.8 after being tested as specified in 4.4.7.3.

3.11.4 High temperature. The test set shall withstand subjection to 160 + 5°F without damage and shall meet the requirements of 3.5 while at 160 ± 5°F. The test shall be conducted as specified in 4.4.7.4.

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3.11.5 Low temperature. The test set shall withstand subjection to -80 + 5°F without damage and shall meet the requirements of 3.5 and 3.8 while at -65 + 5°F. The test shall be performed in accordance with 4.4.7.5.

3.12 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460(MU).

3.13 Workmanship.

3.13.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.13.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.13.3 Burrs. Parts shall be free from burrs.

3.13.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.13.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.13.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 6 complete assemblies (see dwgs 8821412 and 8849096) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests shall be conducted in order specified in Table VI.

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Table VI

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Starting torque (see 3.5)		X
3. Insulation resistance (see 3.6)		X
4. Continuity (see 3.7)		X
5. Performance (see 3.8)		X
6. Diode current leakage (see 3.9)		X
7. *Life (see 3.10)		X
8. Transportation vibration (see 3.11.1)	X	
9. Acceleration (see 3.11.2)		X
10. Transportation shock (see 3.11.3)		X
11. **High temperature (see 3.11.4)		X
12. **Low temperature (see 3.11.5)		X

*Life cycle test 200 cycles each condition at ambient temperature.
**Life cycle test 200 cycles each condition.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VII and Table VIII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table VII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-21	18	35-36	26	70-76	34	245-334	42
[illegible]	9	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table VIII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Test set, atomic weapon T-4133 and test set, atomic weapon, training T-4162 (see dwgs 8821412 and 8849096 prior to potting).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:

101.	Any component part missing, damaged or loose	Visual/Manual
102.	Wires in tension, inadequate or damaged insulation	Visual
103.	Wires routed against sharp edges	Visual
104.	Improper soldering	Visual
105.	Varistor or diode improperly located so it is in contact with a terminal or bracket	Visual

Minor:

201.	Presence of chips or foreign matter unless removable	Visual
202.	Evidence of poor workmanship (see 3.13)	Visual

4.3.2.1.2 Test set, atomic weapon T-4133 and test set, atomic weapon, training, T-4162 (see dwgs 8821412 and 8849096).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
*101.	Height of pin(s) (pins of the same diameter must all appear to be within 1/32 inch in height relative to the adjacent pins)	Visual
102.	Improper location and alignment of the 4 mounting holes and the generator shaft	Gage
103.	Pitch diameter thread in the 4 mounting holes	Gage
104.	Improper cable length	Gage
105.	Any component part missing, loose or damaged	Visual/Manual
106.	Deformation of connector sufficient to impair mating	Visual/Manual
107.	Inadequate potting of assembly or connector	Visual
108.	Inadequate or damaged lead wires or insulation	Visual
109.	Wrong connector	Visual
110.	Foreign matter on terminals	Visual
111.	Portion of connector broken away (Minor chips, scratches, burrs, or gouges which do not affect mating are not objectionable)	Visual

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112. Permanently damaged terminal(s) (gouged, nicked or flattened).....Visual
113. Any pin missing (as evidenced by empty hole in insert).....Visual
114. Any pin bent.....Visual
115. Objective evidence relating to material missing or improper (see 3.1).....Visual
116. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Presence of chips or foreign matter unless removable.....Visual
202. Evidence of poor workmanship (see 3.13).....Visual

*In the event pin(s) do not appear to be within 1/32 inch in height relative to the adjacent pins, Gage (EL-8842555) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.032 inch max from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface of all measurements.

4.3.2.1.3 Carton, packing, ammunition, for control assembly prefire (see dwg 8822039).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Wrong size bag.....Visual
102. Bag not sealed.....Visual
103. Air in bag not minimized before sealing.....Visual
104. Marking improper, illegible, smeared, missing or in wrong location.....Visual
105. Packing material visibly wet.....Visual
106. Incorrect or insufficient cushioning material.....Visual

Minor:
201. General damage.....Visual
202. Evidence of poor packing.....Visual
203. Damage to preservative.....Visual
204. Evidence of poor taping.....Visual
205. Evidence of poor workmanship (see 3.13).....Visual

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4.3.2.1.4 Container, packing, ammunition, for domestic (CONUS) shipments of special weapons items, packing and marking (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Marking missing, mislocated or unidentifible	Visual
	102. Marking improper or smeared	Visual
	103. Packing material visibly wet	Visual
	104. Incorrect or insufficient cushioning material	Visual
Minor:		
	201. General damage	Visual
	202. Evidence of poor packing	Visual
	203. Evidence of poor workmanship (see 3.13)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Starting torque	Major
b.	Insulation resistance	Major
c.	Continuity	Major
d.	Performance	Major
e.	Diode current leakage	Major
f.	Transportation vibration	Major
g.	Low temperature	Major
h.	High temperature	Major

Of these tests, f is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Starting torque, insulation resistance, continuity, performance, and diode current leakage. Each test set shall be tested in accordance with 4.4.1 through 4.4.5. If any item fails to comply with 3.5 through 3.9, the item shall be rejected and removed from the lot.

4.3.3.2 Transportation vibration, high temperature and low temperature. A sample shall be selected at random in accordance with Table IX. The sample shall be tested in accordance with 4.4.7.1, 4.4.7.4 and 4.4.7.5. If any item fails to comply with 3.11.1, 3.11.4 and 3.11.5, the lot shall be rejected.

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Table IX

Lot Size	Sample Size
2-180	2
181-500	3
501-800	5
801-1300	7

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842555). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Starting torque. The test set shall be mounted on a rigid fixture during this test. Measure the torque required to start the generator shaft turning. Record this torque required for each of five trials, making certain the shaft comes to rest after and before each trial.

4.4.2 Insulation resistance. An insulation resistance tester, capable of supplying the required voltage and with which a resistance in megohms can be obtained, shall be used to measure the resistance between the pins specified in Table II. The current shall be limited to a maximum of five milliamperes to protect human life.

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4.4.3 Continuity. Using a suitable 4 terminal instrument, measure the circuit resistance as described.

4.4.4 Performance. Mount the test set on a rigid test fixture and connect to test circuit. Conduct the test using a generator shaft speed of 70 ± 5 revolutions per minute (rpm). The maximum continuous operating period shall be limited to a time of 10 seconds. Conduct the test in accordance with Table III.

4.4.5 Diode current leakage. Measure the leakage current of the diode using an instrument capable of supplying the proper direct current (dc) voltage and a means of reading dc current. Apply voltage to the pins with the polarities specified in Table IV.

4.4.6 Life. Mount the assembly to a fixture suitable for life testing. Provide a closed continuity between pin P1-1 and P1-3. Cycle the assembly by operating according to the following schedule:

1. Cw rotation of shaft at 70 ± 10 rpm for 5 ± 2 seconds.
2. Off for 55 ± 2 seconds.
3. Ccw rotation of shaft at 70 ± 10 rpm for 5 ± 2 seconds.
4. Off for 55 ± 2 seconds.

(This schedule constitutes one cycle.)
During this test the assembly shall be tested in accordance with 4.4.7.4 and 4.4.7.5 to the extent specified in Table VI. At the completion of the 600th cycle the assembly shall be tested in accordance with 4.4.1 through 4.4.5.

4.4.7 Environmental testing (see 3.11).

4.4.7.1 Transportation vibration. The test set shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes, with one axis being along the direction of forward motion (see Figure 1). Vibrate the assembly at the frequencies, amplitudes, and for the times specified in Table V in each of the three major mutually perpendicular axes. Following this test the assembly shall be tested as specified in 4.4.1 through 4.4.4.

4.4.7.2 Acceleration. Mount the test set on the centrifuge so that the direction of forward motion of the assembly is oriented to extend in the radial direction of the centrifuge. Bring the centrifuge up to the rotational speed required to produce the acceleration of 40 ± 5 g's in the direction of forward motion of the assembly. Subject the assembly to 40 ± 5 g's in the direction of forward motion for a period of 5 seconds. Mount the assembly on the direction perpendicular to the radial direction of the centrifuge. Bring the centrifuge up to rotational speed required to produce 18 ± 1 g in the forward direction for a period of 15 seconds.

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4.4.7.3 Transportation shock. The test set shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes. The assembly shall be subjected three times to shocks of 75 ± 8 g's with a half amplitude duration of 9 ± 3 milliseconds in each of the three mutually perpendicular axes. Following this test the assembly shall be tested as specified in 4.4.1 through 4.4.4.

4.4.7.4 High temperature. The test set shall be placed in the test chamber and the tamperature of the chamber raised to 160 ± 5°F. The item shall soak at this temperature (160 ± 5°F) for a minimum of 4 hours prior to testing. While at 160 ± 5°F the assembly shall be tested as specified in 4.4.1.

4.4.7.5 Low temperature. A container providing protection against frosting of the test set shall be used throughout this test. The assembly shall be placed in the test chamber and the temperature of the chamber lowered to -80 ± 5°F. The item shall soak at this temperature (-80 ± 5°F) for a minimum of 4 hours. The chamber temperature shall then be raised to -65 ± 5°F and maintained for an additional 1 hour period or until stabilization (see 6.4) is reached, whichever is longer. While at -65 ± 5°F the assembly shall be tested as specified in 4.4.1 and 4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The test set and prefire control shall be packaged in accordance with dwgs 8808362 and 8822037 using the specified moisture barrier.

5.1.2 Level C. The test set and prefire control shall be packaged in accordance with dwgs 8808362 and 8822037 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The test set and prefire control shall be packed in accordance with dwgs 8808362 and 8822037.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8808362 and 8822037.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

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6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1671 Amendment 4 dated 30 December 1964.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A061

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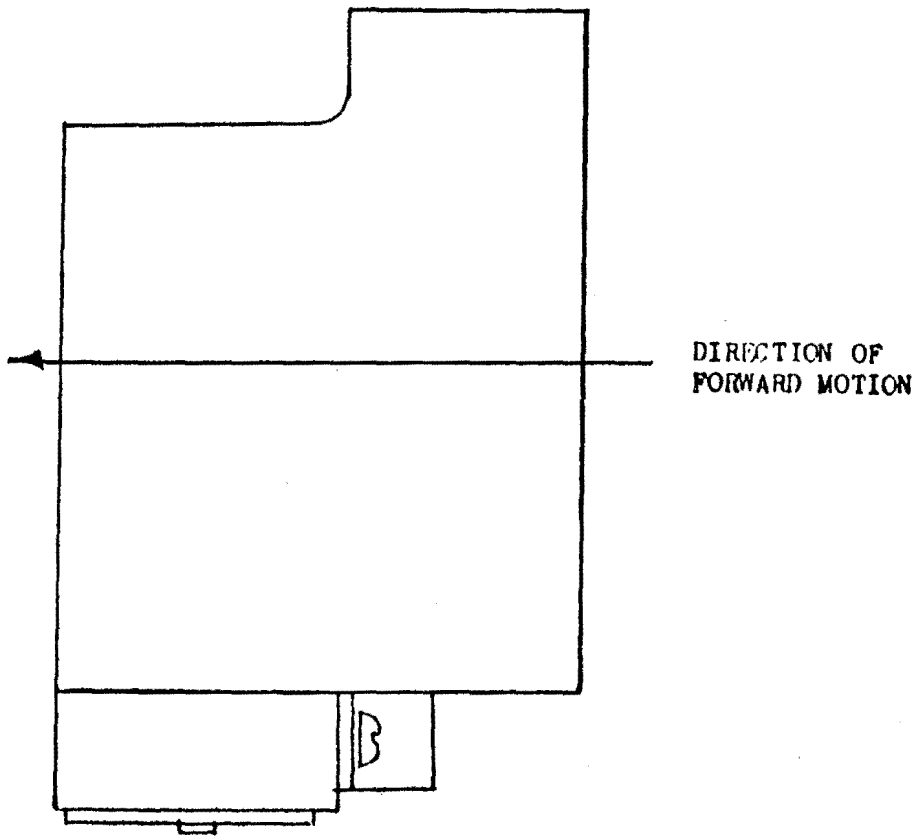


Figure 1

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